

Deep Learning-Based Phishing & Malicious Email Classifier

Deep Learning Applications – ELA-1
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Abstract. This project develops a deep-learning pipeline for automated classification of raw email text into two operational categories: legitimate and phishing/malicious. A synthetic dataset of 2,400 email messages was created to satisfy the minimum dataset requirement while keeping the examples email-oriented and free of personal information. The pipeline performs text cleaning, tokenisation, fixed-length padding/truncation and neural sequence classification using a Bidirectional LSTM (BiLSTM).

Dataset. The dataset contains 1,200 phishing messages and 1,200 legitimate messages. Phishing examples contain patterns such as urgent account warnings, credential verification requests, payment problems and reward claims. Legitimate examples contain routine workplace, appointment, order, support and project communications.

Preprocessing workflow

Stage	Operation
Cleaning	Remove HTML, URLs, unnecessary characters and extra whitespace; lowercase text.
Split	Stratified train/validation/test split: 70% / 15% / 15%.
Tokenisation	Keras TextVectorization converts words into integer token IDs.
Padding	All sequences are padded or truncated to 200 tokens.

The preprocessing is implemented consistently for training and raw-email inference to reduce training-serving skew.

2. Model Architecture and Hyperparameters

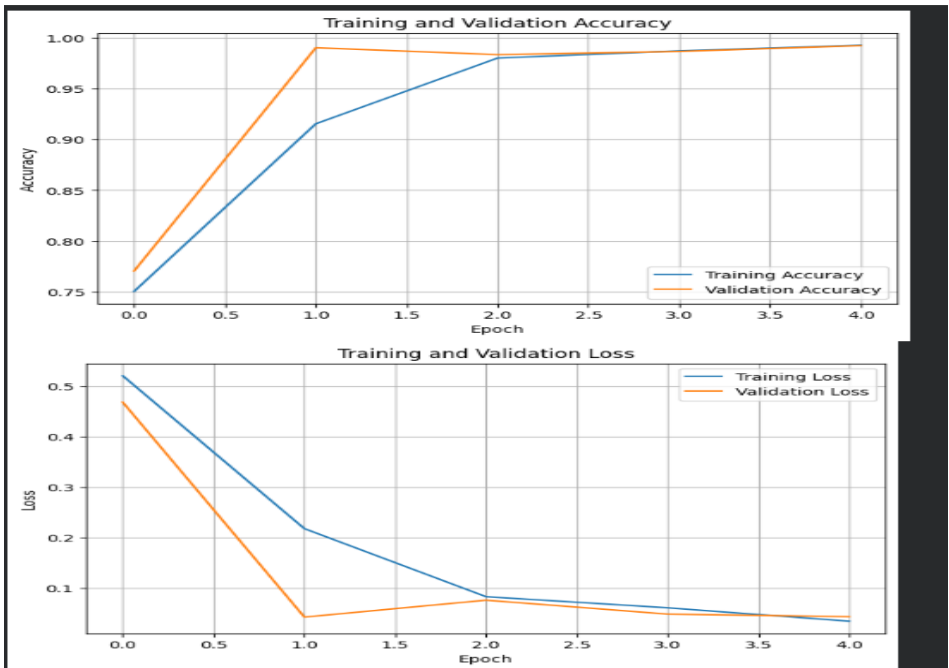
Input	Raw email text
↓	
TextVectorization	Vocabulary $\leq 10,000$; sequence length = 200
↓	
Embedding	128-dimensional learned word embeddings
↓	
Bidirectional LSTM	64 units
↓	
Dropout	0.50
↓	
Dense	32 units, ReLU
↓	
Dropout	0.30
↓	
Output	1 unit, Sigmoid $\rightarrow$ phishing probability

Hyperparameter choices. Adam with a learning rate of 0.001 is used for stable gradient-based optimisation. Binary cross-entropy is appropriate for the two-class sigmoid output. A batch size of 32 and maximum of 15 epochs provide a practical Colab training configuration. Early stopping monitors validation loss and restores the best weights, while ReduceLROnPlateau lowers the learning rate when validation performance stops improving.

Operational flow

Raw email payload $\rightarrow$ cleaning $\rightarrow$ integer sequence $\rightarrow$ fixed-length sequence $\rightarrow$ embedding $\rightarrow$ BiLSTM $\rightarrow$ dense layers $\rightarrow$ phishing probability $\rightarrow$ automated inbox category.

1. Training and Validation Curves



The accuracy curve rises to approximately 99% for both training and validation data, while the loss decreases substantially over the five epochs. The

close final training and validation curves indicate that the model learned the classification patterns effectively without a large final train-validation gap.

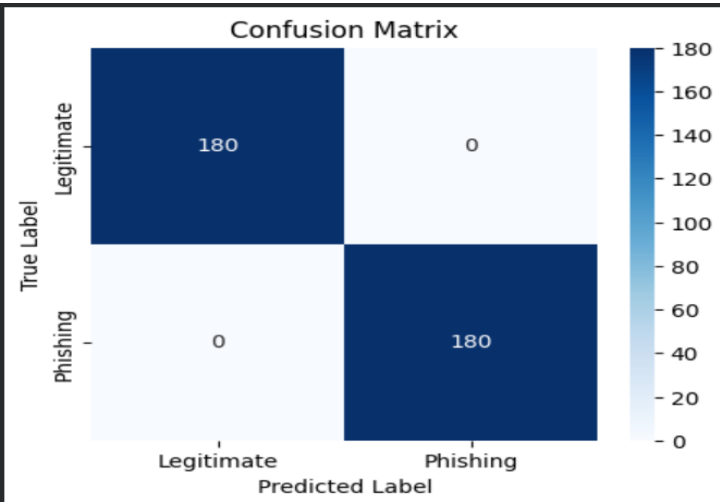
2. Evaluation & Results

Evaluation Metrics

Test Loss:	0.0000
Test Accuracy:	1.0000
Precision:	1.0000
Recall:	1.0000
F1-Score:	1.0000

Classification Report:				
	precision	recall	f1-score	support
Legitimate	1.0000	1.0000	1.0000	180
Phishing	1.0000	1.0000	1.0000	180
accuracy			1.0000	360
macro avg	1.0000	1.0000	1.0000	360
weighted avg	1.0000	1.0000	1.0000	360

Confusion Matrix



Conclusion. The project demonstrates a complete deep-learning workflow for email security classification. The model can consume a raw email text payload and return a phishing probability, making it suitable as a prototype component of an automated inbox-management system. For production use, the model should be retrained and continuously evaluated using a larger organization-specific dataset and additional signals such as sender reputation, domain age, URL reputation, attachment characteristics and authentication headers.